

Abdulaziz EL Sabbagh

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PROFESSIONAL SUMMARY

Computer vision engineer with a research background in aerial perception systems and a track record of taking vision models from prototype to production. Honours research at UTS focused on drone-based spatial detection — curating aerial image datasets, evaluating YOLOv8-Seg, Mask R-CNN, and U-Net, and deploying a live monitoring system. Won the UTS AI Showcase with [InkVision](#), a production CNN-based app with 350+ real-world downloads. Holds an Aeronautical Remote Operators Certificate (AROC) with hands-on experience collecting and processing aerial sensor data. Strong Python and deep learning fundamentals. Built to work end-to-end — from raw sensor input through to deployed systems that reach real users.

CORE COMPETENCIES

- Computer Vision & Object Detection
- Aerial & Drone-Based Perception
- Image Segmentation & Classification
- Sensor Data Processing & Analysis
- Deep Learning Model Deployment
- Prototype-to-Production Systems
- Multi-Architecture Model Benchmarking
- Python & PyTorch Engineering

WORK EXPERIENCE

edFlow Mid 2024 – Present

Co-Founder & Product Lead

Sydney, NSW

- Led end-to-end product delivery** across frontend, backend, and cloud infrastructure — from initial prototype through to a production system used by real customers.
- Built and maintained systems using Next.js, Firebase, Google Cloud, serverless functions, and CI/CD pipelines; mentored 4 interns and translated business requirements into technical delivery plans.
- Drove feature prioritisation and iteration based on direct user and commercial feedback — high ownership, no bureaucracy.

UAVAIR Jan 2019 – Dec 2019

Remote Pilot & Aerial Data Analyst

Sydney, NSW

- Supported commercial drone operations** including pre-flight planning, sensor data collection, and aerial data analysis in real-world environments.
- Delivered pilot training and ensured full regulatory compliance — AROC certified, strong understanding of UAV sensor systems and safe operation protocols.

University of Technology Sydney May 2024 – Present

Student Ambassador (AI Focus)

Sydney, NSW

- Communicate practical AI systems to diverse audiences — image recognition, spatial detection, real-world applications beyond chatbots.

PROJECTS

Honours Project — Drone-Based AI Bushfire Detection System Nov 2024

Designed and built a full aerial perception pipeline for bushfire and smoke detection using drone imagery. Curated and annotated large aerial image datasets. Evaluated and benchmarked YOLOv8-Seg, Mask R-CNN, and U-Net across detection accuracy, segmentation quality, and real-world robustness. Deployed a live web app for monitoring, alerts, and emergency response workflows.

Python · PyTorch · YOLOv8-Seg · Mask R-CNN · U-Net · OpenCV · Aerial Imaging · Dataset Annotation

InkVision — Production Computer Vision iOS App Sep 2023 · UTS AI Showcase Winner

Won the UTS AI Showcase with a virtual tattoo try-on app using real-time forearm detection. Trained a CNN on 1,000+ manually annotated images and shipped a production iOS app that reached 350+ downloads — full pipeline from data collection to deployed product.

Python · CoreML · CNN · iOS (Swift) · Image Segmentation · Dataset Curation

Traffic Sign Detection — Multi-Architecture Benchmarking Jul 2023

Built and benchmarked traffic sign detection models across YOLO v3, v5, and v8 on a 900-image multi-class dataset. Evaluated performance metrics, optimised detection accuracy across four sign categories, and documented architectural trade-offs.

Python · YOLOv3 · YOLOv5 · YOLOv8 · OpenCV · Image Classification · Performance Benchmarking

EDUCATION

Bachelor of Computing Science (Honours) — University of Technology Sydney

Feb 2020 – Nov 2024

Major: Artificial Intelligence and Data Analytics · GPA: 6.0/7.0 · WAM: 81

CERTIFICATIONS

- Aeronautical Remote Operators Certificate (AROC)
- Certificate III in Remote Pilot (VLOS)
- Certificate in Machine Learning using SAS Viya
- Weights & Biases — Building LLM-Powered Apps
- First Aid Certificate · Working With Children Check (NSW)

TECHNICAL SKILLS

- CV & Perception:** OpenCV, YOLOv8, YOLO v3/v5, Mask R-CNN, U-Net, Faster R-CNN, CNNs, GANs, OpenPose
- Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn **Languages:** Python, JavaScript, TypeScript, Swift, SQL, Java
- Data:** Pandas, Matplotlib, dataset annotation, image preprocessing **Cloud & Infra:** GCP, AWS, Firebase, serverless, CI/CD, Vercel
- Systems:** UAV/drone operations, self-hosting, Tailscale